SECTION 2.6

9, (10). $(3x^2 - y^2)dx + (xy - x^3y^{-1})dy = 0$

We begin by placing the equation in standard form.

$$\frac{dy}{dx} = -\frac{3x^2 - y^2}{xy - x^3y^{-1}}$$

$$\frac{dy}{dx} = \frac{-3x^2y + y^3}{xy^2 - x^3}$$

$$\frac{dy}{dx} = \frac{-3x^2y + y^3}{xy^2 - x^3} \cdot \frac{x^{-3}}{x^{-3}}$$

$$\frac{dy}{dx} = \frac{-3y/x + y^3/x^3}{y^2/x^2 - 1}.$$

Now substitute $v = y/x$ or $vx = y$ so that $v + xv' = y'$ and the above equation becomes

$$v + xv' = \frac{-3v + v^3}{v^2 - 1}$$

$$xv' = \frac{-3v + v^3}{v^2 - 1} - v = \frac{-3v + v^3}{v^2 - 1} - \frac{v^3 - v}{v^2 - 1}$$

$$xv' = \frac{-2v}{v^2 - 1}$$

$$\frac{v^2 - 1}{-2v}\ dv = \frac{1}{x}\ dx$$

$$\int \frac{-v}{2} + \frac{1}{2v}\ dv = \int \frac{1}{x}\ dx + C$$

$$\int \frac{-v}{2} + \frac{1}{2v}\ dv = \int \frac{1}{x}\ dx + C$$

$$\frac{-v^2}{4} + \frac{1}{2}\ln|v| = \ln|x| + C.$$

Substituting $v = y/x$ yields

$$\frac{-y^2}{4x^2} + \frac{1}{2}\ln|y/x| = \ln|x| + C$$

$$\frac{-y^2}{4x^2} + \frac{1}{2}\ln|y/x| - \ln|x| = C$$

$$\frac{-y^2}{4x^2} + \frac{1}{2}\ln|y/x| - \frac{2}{2}\ln|x| = C$$
$$\frac{-y^2}{4x^2} + \frac{1}{2}\ln|y/x| - \frac{1}{2}\ln|x^2| = C$$
$$\frac{-y^2}{4x^2} + \frac{1}{2}\ln|y/x^3| = C$$
$$\frac{-y^2}{x^2} + 2\ln|y/x^3| = C$$
$$\frac{-y^2}{x^2} + \ln(y^2/x^6) = C.$$

11, (11). $(y^2 - xy)dx + x^2dy = 0$

We begin by placing the equation in standard form.

$$\frac{dy}{dx} = -\frac{y^2 - xy}{x^2}$$
$$\frac{dy}{dx} = -(y/x)^2 + (y/x).$$

Now substitute $v = y/x$ or $vx = y$ so that $v + xv' = y'$ and the above equation becomes

$$v + xv' = -v^2 + v$$
$$xv' = -v^2$$
$$-v^{-2}\ dv = x^{-1}\ dx$$
$$\int -v^{-2}\ dv = \int x^{-1}\ dx + C$$
$$v^{-1} = \ln|x| + C.$$

Substituting $v = y/x$ yields

$$(y/x)^{-1} = \ln|x| + C$$
$$x/y = \ln|x| + C$$
$$y = \frac{x}{\ln|x| + C}.$$

In addition to the above, the constant function $y = 0$ is also a solution since it satisfies

$$\frac{dy}{dx} = -\frac{y^2 - xy}{x^2}.$$

13, (13). $\frac{dx}{dt} = \frac{x^2+t\sqrt{t^2+x^2}}{tx}$

We begin by placing the equation in standard form.

$$\frac{dx}{dt} = \frac{x}{t} + \frac{\sqrt{t^2+x^2}}{x}$$

$$\frac{dx}{dt} = \frac{x}{t} + \frac{\sqrt{t^2+x^2}}{x} \cdot \frac{1/t}{1/t}$$

$$\frac{dx}{dt} = \frac{x}{t} + \frac{\sqrt{1+(x/t)^2}}{x/t}$$

Now substitute $v = x/t$ or $vt = x$ so that $v + tv' = x'$ and the above equation becomes

$$v + tv' = v + \frac{\sqrt{1+v^2}}{v}$$

$$tv' = \frac{\sqrt{1+v^2}}{v}$$

$$\frac{v}{\sqrt{1+v^2}}\ dv = t^{-1}\ dt$$

$$\int \frac{v}{\sqrt{1+v^2}}\ dv = \int\ t^{-1}\ dt + C$$

$$\sqrt{1+v^2} = \ln|t| + C.$$

(Note: to integrate one may wish to use the substitution $u = 1 + v^2$)

Substituting $v = x/t$ yields

$$\sqrt{1+(x/t)^2} = \ln|t| + C.$$

15, (15). $\frac{dy}{dx} = \frac{x^2-y^2}{3xy}$

We begin by placing the equation in standard form.

$$\frac{dy}{dx} = \frac{x}{3y} - \frac{y}{3x}$$

$$\frac{dy}{dx} = \frac{1}{3}\left[\frac{1}{y/x} - \frac{y}{x}\right]$$

Now substitute $v = y/x$ or $vx = y$ so that $v + xv' = y'$ and the above equation becomes

$$v + xv' = \frac{1}{3}\left[\frac{1}{v} - v\right]$$

$$xv' = \frac{1}{3}\left[\frac{1}{v} - 4v\right]$$

$$xv' = \frac{1}{3}\left[\frac{1-4v^2}{v}\right]$$

$$\frac{v}{1-4v^2}\ dv = \frac{1}{3x}\ dx$$

$$\int \frac{v}{1-4v^2}\ dv = \int \frac{1}{3x}\ dx + C$$

$$-\frac{1}{8}\ln|1-4v^2| = \frac{1}{3}\ln|x| + C.$$

(Note: to integrate one may wish to use the substitution $u = 1 - 4v^2$)

Substituting $v = y/x$ yields

$$-\frac{1}{8}\ln|1-4(y/x)^2| = \frac{1}{3}\ln|x| + C$$

$$-\frac{1}{8}\ln|1-4(y/x)^2| - \frac{1}{3}\ln|x| = C$$

$$\frac{1}{8}\ln|1-4(y/x)^2| + \frac{1}{3}\ln|x| = C$$

$$3\ln|1-4(y/x)^2| + 8\ln|x| = C$$

$$\ln|1-4(y/x)^2|^3 + \ln|x|^8 = C$$

$$\ln[|1-4(y/x)^2|^3|x|^8] = C$$

$$|1-4(y/x)^2|^3|x|^8 = C,\ C > 0$$

$$(1-4(y/x)^2)^3x^8 = C,\ |C| > 0$$

$$(1-4(y/x)^2)^3(x^2)^3x^2 = C,\ |C| > 0$$

$$(x^2-4y^2)^3x^2 = C,\ |C| > 0.$$

17, (17). $\frac{dy}{dx} = \sqrt{x+y} - 1$

Since the equation is already in standard form we perform the substitution $v = x + y$ so that $v' = 1 + y'$ or $y' = v' - 1$ and obtain

$$v' - 1 = \sqrt{v} - 1$$

$$v' = \sqrt{v}$$

$$v^{-1/2}\ dv = 1\ dx$$

$$\int\ v^{-1/2}\ dv = \int\ 1\ dx + C$$

$$2v^{1/2} = x + C.$$

Substituting $v = y + x$ yields

$$2\sqrt{x+y} = x + C$$

$$4(x+y) = (x+C)^2$$

$$y = (x+C)^2/4 - x.$$

It is not obvious that $y = -x$ is also a solution to the equation; that this is true can be seen since $y' = \frac{d}{dx}\ (-x) = -1$ and substituting $y = -x$ into the right hand side of the original equation also yields -1. This particular solution could not be obtained from the general solution since

$$-x = (x+C)^2/4 - x$$

simplifies to

$$0 = (x+C)^2/4;$$

This last equation is true only if $x = -C$ which is not possible since C is a constant.

19, (19). $\frac{dy}{dx} = (x - y + 5)^2$

Since the equation is already in standard form we perform the substitution $v = x - y$ so that $v' = 1 - y'$ or $y' = 1 - v'$ and obtain

$$1 - v' = (v+5)^2$$

$$v' = 1 - (v+5)^2$$

$$\frac{1}{1-(v+5)^2}\ dv = 1\ dx$$

$$\int\ \frac{1}{1-(v+5)^2}\ dv = \int\ 1\ dx.$$

To evaluate the integral we substitute $z = v + 5$ and obtain

$$\int \frac{1}{1-(v+5)^2}\,dv = \int \frac{1}{1-z^2}\,dz = \int \frac{1}{(1-z)(1+z)}\,dz = \int \frac{A}{1-z} + \frac{B}{1+z}\,dz.$$

Solving for A and B gives

$$\frac{1}{1-z^2} = \frac{A}{1-z} + \frac{B}{1+z}$$

$$\frac{1}{1-z^2} = \frac{A(1+z)}{(1-z)(1+z)} + \frac{B(1-z)}{(1-z)(1+z)}$$

so that

$$1 = A(1+z) + B(1-z).$$

Choosing $z = 1$ gives $A = 1/2$ and choosing $z = -1$ gives $B = 1/2$; hence

$$\int \frac{1}{1-z^2}\,dz = (1/2)\int \frac{1}{1-z} + \frac{1}{1+z}\,dz$$

$$\int \frac{1}{1-z^2}\,dz = (1/2)(-\ln|1-z| + \ln|1+z|) = (1/2)\ln|(1+z)/(1-z)|$$

so that

$$\int \frac{1}{1-(v+5)^2}\,dv = (1/2)\ln|(v+6)/(-v-4)|$$

$$(1/2)\ln|(v+6)/(-v-4)| = x + C.$$

Substituting $v = x - y$ yields

$$(1/2)\ln|(x-y+6)/(-x+y-4)| = x + C$$

$$\ln|(x-y+6)/(-x+y-4)| = 2x + C$$

$$|(x-y+6)/(-x+y-4)| = e^{2x+C} = e^{2x}e^{C} = Ce^{2x},\ C > 0$$

$$(x-y+6)/(-x+y-4) = Ce^{2x},\ |C| > 0$$

$$(x-y+6) = (-x+y-4)Ce^{2x}$$

$$x - y + 6 = -(x+4)Ce^{2x} + yCe^{2x}$$

$$y(Ce^{2x}+1) = (x+4)Ce^{2x} + x + 6 = (x+4)(Ce^{2x}+1) + 2$$

$$y = (x+4) + 2/[Ce^{2x}+1]$$

$$y = x + 4[Ce^{2x}+1]/[Ce^{2x}+1] + 2/[Ce^{2x}+1]$$

$$y = x + [4Ce^{2x}+6]/[Ce^{2x}+1].$$

It is not obvious that $y = x + 4$ is also a solution to the equation; that this is true can be seen since $y' = \frac{d}{dx}\,(x+4) = 1$ and substituting $y = x + 4$

into the right hand side of the original equation also yields 1. This particular solution could not be obtained from the general solution since

$$x + 4 = x + [4Ce^{2x} + 6]/[Ce^{2x} + 1]$$

simplifies to

$$\begin{aligned} 4 &= [4Ce^{2x} + 6]/[Ce^{2x} + 1] \\ 4[Ce^{2x} + 1] &= [4Ce^{2x} + 6] \\ 4 &= 6. \end{aligned}$$

This last equation is clearly not true.

21, (21). $\frac{dy}{dx} + \frac{y}{x} = x^2y^2$

Since the equation is already in standard form we note $n = 2$ and perform the substitution $v = y^{1-n} = y^{1-2} = y^{-1}$ so that $v' = -y^{-2}y'$ or $y' = -y^2v'$ and obtain

$$\begin{aligned} -y^2v' + \frac{y}{x} &= x^2y^2 \\ v' - \frac{1}{x}y^{-1} &= -x^2 \\ v' - \frac{1}{x}v &= -x^2. \end{aligned}$$

We obtain the integrating factor $\mu(x) = e^{\int -\frac{1}{x}\,dx} = e^{-\ln|x|} = |x|^{-1}$ so that

$$\begin{aligned} (v|x|^{-1})' &= -x^2|x|^{-1} \\ (vx^{-1})' &= -x^2x^{-1} = -x \\ vx^{-1} &= \int -x\,dx + C = -x^2/2 + C \\ v &= -x^3/2 + Cx. \end{aligned}$$

Substituting $v = y^{-1}$ gives

$$\begin{aligned} y^{-1} &= -x^3/2 + Cx \\ y &= 1/(-x^3/2 + Cx) \\ y &= 2/(Cx - x^3). \end{aligned}$$

A second solution to the original equation is given by $y = 0$. This can not be obtained from the general solution since the equation

$$0 = 2/(Cx - x^3)$$

can not be satisfied by any real C.

23, (23). $\frac{dy}{dx} = \frac{2y}{x} - x^2y^2$

Placing the problem in standard form yields

$$\frac{dy}{dx} - \frac{2y}{x} = -x^2y^2$$

We note $n = 2$ and perform the substitution $v = y^{1-n} = y^{1-2} = y^{-1}$ so that $v' = -y^{-2}y'$ or $y' = -y^2v'$ and obtain

$$-y^2v' - \frac{2y}{x} = -x^2y^2$$

$$v' + \frac{2}{x}y^{-1} = x^2$$

$$v' + \frac{2}{x}v = x^2.$$

We obtain the integrating factor $\mu(x) = e^{\int \frac{2}{x}\, dx} = e^{2\ln|x|} = |x|^2 = x^2$ so that

$$(vx^2)' = x^2x^2$$

$$(vx^2)' = x^4$$

$$vx^2 = \int x^4\, dx + C = x^5/5 + C$$

$$v = x^3/5 + C/x^2.$$

Substituting $v = y^{-1}$ gives

$$y^{-1} = x^3/5 + C/x^2$$

$$y = 1/(x^3/5 + C/x^2)$$

$$y = 5x^2/(x^5 + C).$$

A second solution to the original equation is given by $y = 0$. This can not be obtained from the general solution since the equation

$$0 = 5x^2/(x^5 + C)$$

can not be satisfied by any real C.

25, (25). $\frac{dx}{dt} + tx^3 + x/t = 0$

Placing the problem in standard form yields

$$\frac{dx}{dt} + x/t = -tx^3.$$

We note $n = 3$ and perform the substitution $v = x^{1-n} = x^{1-3} = x^{-2}$ so that $v' = -2x^{-3}x'$ or $x' = -x^3v'/2$ and obtain

$$-x^3v'/2 + x/t = -tx^3$$

$$v' - 2x^{-2}/t = 2t$$

$$v' - 2v/t = 2t.$$

We obtain the integrating factor $\mu(t) = e^{\int \frac{-2}{t}\, dt} = e^{-2\ln|t|} = |t|^{-2} = t^{-2}$ so that

$$(vt^{-2})' = (2t)(t^{-2})$$

$$(vt^{-2})' = 2/t$$

$$vt^{-2} = \int 2/t\ dt + C$$

$$vt^{-2} = 2\ln|t| + C$$

$$v = t^2(2\ln|t| + C).$$

Substituting $v = x^{-2}$ gives

$$x^{-2} = t^2(2\ln|t| + C)$$

$$x^{-2} = 2t^2\ln|t| + Ct^2.$$

A second solution to the original equation is given by $x = 0$. This can not be obtained from the general solution since the equation

$$0 = 1/\sqrt{2t^2\ln|t| + Ct^2}$$

can not be satisfied by any real C.

27, (27). $\frac{dr}{d\theta} = \frac{r^2+2r\theta}{\theta^2}$

Placing the problem in standard form yields

$$\frac{dr}{d\theta} - \frac{2r}{\theta} = \frac{r^2}{\theta^2}$$

We note $n = 2$ and perform the substitution $v = r^{1-n} = r^{1-2} = r^{-1}$ so that $v' = -r^{-2}r'$ or $r' = -r^2 v'$ and obtain

$$-r^2 v' - \frac{2r}{\theta} = \frac{r^2}{\theta^2}$$

$$v' + \frac{2}{\theta} r^{-1} = -\frac{1}{\theta^2}$$

$$v' + \frac{2}{\theta} v = -\frac{1}{\theta^2}.$$

We obtain the integrating factor $\mu(\theta) = e^{\int \frac{2}{\theta}\, d\theta} = e^{2\ln|\theta|} = |\theta|^2 = \theta^2$ so that

$$(v\theta^2)' = -\frac{1}{\theta^2}\theta^2$$

$$(v\theta^2)' = -1$$

$$v\theta^2 = \int -1 \; d\theta + C$$

$$v\theta^2 = -\theta + C$$

$$v = -1/\theta + C/\theta^2.$$

Substituting $v = r^{-1}$ gives

$$r^{-1} = -1/\theta + C/\theta^2 = (C - \theta)/\theta^2$$

$$r = \theta^2/(C - \theta).$$

A second solution to the original equation is given by $r = 0$. This can not be obtained from the general solution since the equation

$$0 = \theta^2/(C - \theta)$$

can not be satisfied by any real C.